A PROPER ALGEBRAIC-SPACE MORPHISM HAS A FINITE UNIVERSALLY INJECTIVE KERNEL-PAIR COEQUALIZER

THE STACKS PROJECT AUTHORS

Editorial note. Public280 concerns constructible etale sheaf presentations and public293 is Beck monadicity. Neither supplies the proper algebraic-space kernel-pair coequalizer or its finite radicial factorization.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Locally Noetherian algebraic spaces, proper coherent pushforward, relative Spec, Stein factorization and flat base change. The local etale-check and proper-surjective descent support are retained; the coequalizer itself is not assumed.

Source reference. Original section 0G2U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20069–20069. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03FX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4508–4508. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GFQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20520–20520. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 0ADG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13089–13089. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 0ADQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13098–13098. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 03EB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4450–4450. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0GFW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20526–20526. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GF6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20501–20501. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04WP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6367–6367. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04XY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6411–6411. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08AR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10541–10541. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 081W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10231–10231. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 081X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10232–10232. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6095–6095. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 0A1B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12664–12664. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00GQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 616–616. Stable Stacks tag for an expressly allowed

foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 03MV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4763–4763. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 040X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5272–5272. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03MH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4749–4749. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03MG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4748–4748. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04ZK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6468–6468. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 073K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9064–9064. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05Z1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7675–7675. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let S be a scheme. Let*

$$g : Y \longrightarrow X$$

be a morphism of algebraic spaces over S. Assume X is locally Noetherian, and g is proper. Let $R = Y \times_X Y$ with projection morphisms $t, s : R \to Y$. There exists a coequalizer X' of $s, t : R \to Y$ in the category of algebraic spaces over S. Moreover

1. *The morphism $X' \to X$ is finite.*
2. *The morphism $Y \to X'$ is proper.*
3. *The morphism $Y \to X'$ is surjective.*
4. *The morphism $X' \to X$ is universally injective.*
5. *If g is surjective, the morphism $X' \to X$ is a universal homeomorphism.*

Lemma 0.2. *Let S be a scheme. Let B be an algebraic space over S. Let $\mathcal{I} \to (Sch/S)_{fppf}$, $i \mapsto X_i$ be a diagram of algebraic spaces over B. Let $(X, X_i \to X)$ be a cocone for the diagram in the category of algebraic spaces over B (Categories, Remark 0G2U). Assume that*

(1) *the base change functor $X_{spaces,étale} \to \lim X_{i,spaces,étale}$, sending U to $U_i = X_i \times_X U$ is an equivalence,*
(2) *given*
   (a) *B' affine and étale over B,*
   (b) *Z an affine scheme over B',*
   (c) *$U \to X \times_B B'$ an étale morphism of algebraic spaces with U affine,*
   (d) *$f_i : U_i \to Z$ a cocone over B' of the diagram $i \mapsto U_i = U \times_X X_i$,*

   there exists a unique morphism $f : U \to Z$ over B' such that f_i equals the composition $U_i \to U \to Z$.

Then $X = \operatorname{colim} X_i$ in the category of all algebraic spaces over B.

Proof. In this paragraph we reduce to the case where B is an affine scheme. Let $B' \to B$ be an étale morphism of algebraic spaces. Observe that conditions (1) and (2) are preserved if we replace B, X_i, X by B', $X_i \times_B B'$, $X \times_B B'$. Let $\{B_a \to B\}_{a \in A}$ be an étale covering with B_a affine, see Properties of Spaces, Lemma 03FX. For $a \in A$ denote X_a, $X_{a,i}$ the base changes of X and the diagram to B_a. For $a, b \in A$ denote $X_{a,b}$ and $X_{a,b,i}$ the base changes of X and the diagram to $B_a \times_B B_b$. By Lemma 0GFQ it suffices to prove that $X_a = \operatorname{colim} X_{a,i}$ and $X_{a,b} = \operatorname{colim} X_{a,b,i}$. This reduces us to the case where $B = B_a$ (an affine scheme) or $B = B_a \times_B B_b$ (a separated scheme). Repeating the argument once more, we conclude that we may assume B is an affine scheme (this uses that the intersection of affine opens in a separated scheme is affine).

Assume B is an affine scheme. Let Z be an algebraic space over B. We have to show

$$\operatorname{Mor}_B(X, Z) \longrightarrow \lim \operatorname{Mor}_B(X_i, Z)$$

is a bijection.

Proof of injectivity. Let $f, g : X \to Z$ be morphisms such that the compositions $f_i, g_i : X_i \to Z$ are the same for all i. Choose an affine scheme Z' and an étale morphism $Z' \to Z$. By Properties of Spaces, Lemma 03FX we know we can cover Z by such affines. Set $U = X \times_{f,Z} Z'$ and $U' = X \times_{g,Z} Z'$ and denote $p : U \to X$ and $p' : U' \to X$ the projections. Since $f_i = g_i$ for all i, we see that

$$U_i = X_i \times_{f_i,Z} Z' = X_i \times_{g_i,Z} Z' = U'_i$$

compatible with transition morphisms. By (1) there is a unique isomorphism $\epsilon : U \to U'$ as algebraic spaces over X, i.e., with $p = p' \circ \epsilon$ which is compatible with the displayed identifications. Choose an étale covering $\{h_a : U_a \to U\}$ with U_a affine. By (2) we see that $f \circ p \circ h_a = g \circ p' \circ \epsilon \circ h_a = g \circ p \circ h_a$. Since $\{h_a : U_a \to U\}$ is an étale covering we conclude $f \circ p = g \circ p$. Since the collection of morphisms $p : U \to X$ we obtain in this manner is an étale covering, we conclude that $f = g$.

Proof of surjectivity. Let $f_i : X_i \to Z$ be an element of the right hand side of the displayed arrow in the first paragraph of the proof. It suffices to find an étale covering $\{U_c \to X\}_{c \in C}$ such that the families $f_{c,i} \in \lim_i \operatorname{Mor}_B(X_i \times_X U_c, Z)$ come from morphisms $f_c : U_c \to Z$. Namely, by the uniqueness proved above the morphisms f_c will agree on $U_c \times_X U_b$ and hence will descend to give the desired morphism $f : X \to Z$. To find our covering, we first choose an étale covering $\{g_a : Z_a \to Z\}_{a \in A}$ where each Z_a is affine. Then we let $U_{a,i} = X_i \times_{f_i,Z} Z_a$. By (1)

we find $U_{a,i} = X_i \times_X U_a$ for some algebraic spaces U_a étale over X. Then we choose étale coverings $\{U_{a,b} \to U_a\}_{b \in B_a}$ with $U_{a,b}$ affine and we consider the morphisms

$$U_{a,b,i} = X_i \times_X U_{a,b} \to X_i \times_X U_a = X_i \times_{f_i, Z} Z_a \to Z_a$$

By (2) we obtain morphisms $f_{a,b} : U_{a,b} \to Z_a$ compatible with these morphisms. Setting $C = \coprod_{a \in A} B_a$ and for $c \in C$ corresponding to $b \in B_a$ setting $U_c = U_{a,b}$ and $f_c = g_a \circ f_{a,b} : U_c \to Z$ we conclude. □

Lemma 0.3. *Let S be a scheme. Let $f : X \to Y$ be a proper surjective morphism of algebraic spaces over S. Any descent datum $(U/X, \varphi)$ relative to f (Descent on Spaces, Definition 0ADG) with U étale over X is effective (Descent on Spaces, Definition 0ADQ). More precisely, there exists an étale morphism $V \to Y$ of algebraic spaces whose corresponding canonical descent datum is isomorphic to $(U/X, \varphi)$.*

Proof. Recall that U gives rise to a representable sheaf $\mathcal{F} = h_U$ in $Sh(X_{spaces,étale}) = Sh(X_{étale})$, see Properties of Spaces, Section 03EB. The descent datum on U relative to f exactly gives a descent datum $(\mathcal{F}, \varphi)$ for étale sheaves with respect to $\{X \to Y\}$. By Lemma 0GFW this descent datum is effective. Let $\mathcal{G}$ be the corresponding sheaf on $Y_{étale}$. By Properties of Spaces, Lemma 0GF6 we obtain an étale morphism $V \to Y$ of algebraic spaces corresponding to $\mathcal{G}$; we omit the verification of the set theoretic condition[1]. The given isomorphism $\mathcal{F} \to f_{small}^{-1}\mathcal{G}$ corresponds to an isomorphism $U \to V \times_Y X$ compatible with the descent datum. □

Proof. Denote $h : R \to X$ denote the composition of either s or t with g. Then h is proper by Morphisms of Spaces, Lemmas 04WP and 04XY. The sheaves

$$g_*\mathcal{O}_Y \quad \text{and} \quad h_*\mathcal{O}_R$$

are coherent $\mathcal{O}_X$-algebras by Cohomology of Spaces, Lemma 08AR. The X-morphisms s, t induce $\mathcal{O}_X$-algebra maps $s^\sharp, t^\sharp$ from the first to the second. Set

$$\mathcal{A} = \text{Equalizer}\left(s^\sharp, t^\sharp : g_*\mathcal{O}_Y \longrightarrow h_*\mathcal{O}_R\right)$$

Then $\mathcal{A}$ is a coherent $\mathcal{O}_X$-algebra and we can define

$$X' = \underline{\text{Spec}}_X(\mathcal{A})$$

as in Morphisms of Spaces, Definition 081W. By Morphisms of Spaces, Remark 081X and functoriality of the $\underline{\text{Spec}}$ construction there is a factorization

$$Y \longrightarrow X' \longrightarrow X$$

and the morphism $g' : Y \to X'$ equalizes s and t.

Before we show that X' is the coequalizer of s and t, we show that $Y \to X'$ and $X' \to X$ have the desired properties. Since $\mathcal{A}$ is a coherent $\mathcal{O}_X$-module it is clear that $X' \to X$ is a finite morphism of algebraic spaces. This proves (1). The morphism $Y \to X'$ is proper by Morphisms of Spaces, Lemma 04NX. This proves (2). Denote $Y \to Y' \to X$ with $Y' = \underline{\text{Spec}}_X(g_*\mathcal{O}_Y)$ the Stein factorization of g, see More on Morphisms of Spaces, Theorem 0A1B. Of course we obtain morphisms $Y \to Y' \to X' \to X$ fitting with the morphisms studied above. Since $\mathcal{O}_{X'} \subset g_*\mathcal{O}_Y$ is a finite extension we see that $Y' \to X'$ is finite and surjective. Some details omitted; hint: use Algebra, Lemma 00GQ and reduce to the affine case by étale localization. Since $Y \to Y'$ is surjective (with geometrically connected fibres) we conclude that $Y \to X'$ is surjective. This proves (3). To show that $X' \to X$ is universally injective, we have to show that $X' \to X' \times_X X'$ is surjective, see Morphisms of Spaces, Definition 03MV and Lemma 040X. Since $Y \to X'$

[1]It follows from the fact that $\mathcal{F}$ satisfies the corresponding condition.

is surjective (see above) and since base changes and compositions of surjective morphisms are surjective by Morphisms of Spaces, Lemmas 03MH and 03MG we see that $Y \times_X Y \to X' \times_X X'$ is surjective. However, since $Y \to X'$ equalizes s and t, we see that $Y \times_X Y \to X' \times_X X'$ factors through $X' \to X' \times_X X'$ and we conclude this latter map is surjective. This proves (4). Finally, if g is surjective, then since g factors through $X' \to X$ we see that $X' \to X$ is surjective. Since a surjective, universally injective, finite morphism is a universal homeomorphism (because it is universally bijective and universally closed), this proves (5).

In the rest of the proof we show that $Y \to X'$ is the coequalizer of s and t in the category of algebraic spaces over S. Observe that X' is locally Noetherian (Morphisms of Spaces, Lemma 04ZK). Moreover, observe that $Y \times_{X'} Y \to Y \times_X Y$ is an isomorphism as $Y \to X'$ equalizes s and t (this is a categorical statement). Hence in order to prove the statement that $Y \to X'$ is the coequalizer of s and t, we may and do assume $X = X'$. In other words, $\mathcal{O}_X$ is the equalizer of the maps $s^\sharp, t^\sharp : g_*\mathcal{O}_Y \to h_*\mathcal{O}_R$.

Let $X_1 \to X$ be a flat morphism of algebraic spaces over S with X_1 locally Noetherian. Denote $g_1 : Y_1 \to X_1$, $h_1 : R_1 \to X_1$ and $s_1, t_1 : R_1 \to Y_1$ the base changes of g, h, s, t to X_1. Of course g_1 is proper and $R_1 = Y_1 \times_{X_1} Y_1$. Since we have flat base change for pushforward of quasi-coherent modules, Cohomology of Spaces, Lemma 073K, we see that $\mathcal{O}_{X_1}$ is the equalizer of the maps $s_1^\sharp, t_1^\sharp : g_{1,*}\mathcal{O}_{Y_1} \to h_{1,*}\mathcal{O}_{R_1}$. Hence all the assumptions we have are preserved by this base change.

At this point we are going to check conditions (1) and (2) of Lemma 0.2. Condition (1) follows from Lemma 0.3 and the fact that g is proper and surjective (because $X = X'$). To check condition (2), by the remarks on base change above, we reduce to the statement discussed and proved in the next paragraph.

Assume $S = \mathrm{Spec}(A)$ is an affine scheme, $X = X'$ is an affine scheme, and Z is an affine scheme over S. We have to show that

$$\mathrm{Mor}_S(X, Z) \longrightarrow \mathrm{Equalizer}(s, t : \mathrm{Mor}_S(Y, Z) \to \mathrm{Mor}_S(R, Z))$$

is bijective. However, this is clear from the fact that $X = X'$ which implies $\mathcal{O}_X$ is the equalizer of the maps $s^\sharp, t^\sharp : g_*\mathcal{O}_Y \to h_*\mathcal{O}_R$ which in turn implies

$$\Gamma(X, \mathcal{O}_X) = \mathrm{Equalizer}\left(s^\sharp, t^\sharp : \Gamma(Y, \mathcal{O}_Y) \to \Gamma(R, \mathcal{O}_R)\right)$$

Namely, we have

$$\mathrm{Mor}_S(X, Z) = \mathrm{Hom}_A(\Gamma(Z, \mathcal{O}_Z), \Gamma(X, \mathcal{O}_X))$$

and similarly for Y and R, see Properties of Spaces, Lemma 05Z1. □

References